



Curriculum *for* **Master of Computer Application**

Admission Batch 2025-2027
(With effective from 2025-2026)

School of Computer Application and Technology

University Campus: Alaniya, Jhalawar Road, Kota-325003 (Rajasthan)
Corporate Office: CP Tower, IPIA, Road No.1, Kota-324005, (Raj.) Ph.:0744-3040045
Email: info@cpuniverse.in | **website:** www.cpuniverse.in
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Subject wise Credit

COURSECODE	COURSENAME	CREDITS	L	T	P
EGL805	Business Writing and Professional Communication	2	2	0	0
CAL831	Software Engineering & CASE Tools	3	3	0	0
CAL817	Operating System	3	3	0	0
CAL813	Advance Database Management System	4	3	0	2
CAL815	Web Technology	4	3	0	2
CAL851	Programming with Python	4	3	0	2
CAL818	Cloud Computing	3	3	0	0

Abbreviations	
L: No of Lecture per week	LC: Theory Papers per Semester
T: No of Tutorial per week	CR: Credit per Subject
P: No of Practical per week	DC: Departmental Core
DE: Departmental Elective	DCL: Departmental Clinical

CAL831 Software Engineering & CASE Tools **3 Credits (3-0-0)**

Syllabus

Unit-1:

Introduction: Introduction to software engineering, Importance of software, The evolving role of software, Software Characteristics, Software Components, Software Applications, Software Crisis, Software engineering problems, Software Development Life Cycle, Software Process.

Unit-2:

Software Requirement Specification: Analysis Principles, WaterFall Model, The Incremental Model, Prototyping, Spiral Model, Role of management in software development, Role of matrices and Measurement, Problem Analysis, Requirement specification, Monitoring and Control.

Software-Design: Design principles, problem partitioning, abstraction, and top down and bottom up design, Structured approach, functional versus object oriented approach, design specifications and verification, Monitoring and control, Cohesion, Coupling, Fourth generation techniques, Functional independence, Software Architecture, Transaction and Transform Mapping, Component – level Design, Fourth Generation Techniques.

Unit-3:

Coding: Top-Down and Bottom –Up programming, structured programming, information hiding, programming style and internal documentation.

Testing: Testing principles, Levels of testing, functional testing, structural testing, test plane, test case specification, reliability assessment, software testing strategies, Verification & validation, Unit testing, Integration Testing, Alpha & Beta testing, system testing and debugging

Unit-4:

Software Project Management: The Management spectrum- (The people, the product, the process, the project), cost estimation, project scheduling, staffing, software configuration management, Structured Vs Unstructured maintenance, quality assurance, project monitoring, risk management.

Unit-5:

Software Reliability & Quality Assurance: Reliability issues, Reliability metrics, Reliability growth modeling, Software quality, ISO 9000 certification for software industry, SEI capability maturity model, comparison between ISO & SEI CMM.

CASE (Computer Aided Software Engineering): CASE and its Scope, CASE support in software life cycle, documentation, project management, Reverse Software Engineering, Architecture of CASE environment.

Suggested Books and References

Text Books :

1. Pressman, Roger S., “Software Engineering: A Practitioner’s Approach”, McGraw Hill
2. Jalote, Pankaj, “Software Engineering”, Narosa
3. Schaum’s Series, “Software Engineering”, TMH

Reference Books

1. Alexis, Leon and Mathews Leon, “Fundamental of Software Engineering”, Vikas
2. Sommerville, Ian, “Software Engineering”, AWL, 2000
3. Rajib Mall, “Fundamentals of Software Engineering”, PHI Publication.
4. Subhajit Datta, “software engineering concepts & Applications”, Oxford Univ. Press.
5. Carlo Ghezzi, “Fundamentals of software engineering” Pearson Prentice Hall.

Suggested E-resources

1. <http://nriit.edu.in/e-resources-cse/>
2. <https://www.javatpoint.com/software-engineering>
3. <https://www.guru99.com/software-engineering-tutorial.html>

CAL817 Operating System 3 Credits (3-0-0)

Syllabus

Unit-1:

Introduction to the Operating System (OS), Types of OS: Batch System, Time Sharing System, Real Time System. Multi Programming, Distributed System, Functions and Services of OS.

Unit-2:

Process Management: Process Concept, Process State, Process Control Block, Process Scheduling, CPU

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Scheduling - CPU Scheduling, Scheduling Criteria, Scheduling Algorithms, Preemptive & Non Preemptive Scheduling.

Unit-3:

Deadlocks-System model, Characterization, Deadlock Prevention, Deadlock Avoidance and Detection, Recovery from deadlock.

Unit-4:

Memory Management: Logical Address, Physical Address Contiguous Allocation, External and Internal Fragmentation

Virtual Memory: Demand paging, page replacement, allocation of frames, thrashing.

Unit-5:

Information Management: File Concept, Access Methods, Directory Structure. Device Management: Disk Structure, Disk Scheduling Algorithms.

Text books:

1. Silberschatz and Galvin," Operating System Concept", Addison Wesley,

Reference books:

1. Tannenbaum,"Operating System Concept", Addison Wesley, 2002.

Suggested E-resources

1. <https://www.javatpoint.com/>
2. <https://nptel.ac.in/courses/106105214>
3. https://www.tutorialspoint.com/operating_system/index.htm

CAL813 Advanced Database Management System

4 Credits (3-0-2)

Syllabus

Unit-1:

Introduction: Elements of Database System, Characteristics of database approach, File system versus DBMS, data models, DBMS architecture and data independence. Role of DBA, DDL, DML and DCL.

E-R Modeling: Entity types, entity set, attribute and key, relationships, relation types, roles and structural constraints, weak entities, enhanced E-R and overview of object modeling. Specialization and generalization.

Unit-2:

Relational Data Model: Relational model concepts: The catalog, base tables and views. Relational Data Objects - Domains and Relations: Domains, relations, kinds of

relations, relations and predicates, relational databases. Relational constraints, relational algebra.

Data Normalization: Functional dependencies, Normal form up to 3rd normal form & BCNF

Unit-3:

SQL: SQL queries, programming using SQL (PL/SQL), Integrity Constraints, Roles and privileges. PL/SQL – Introduction to PL/SQL – Declare, begin statements, Variables, Control Structure, PL/SQL Transactions – Savepoint, Cursor, PL/SQL Database Objects – Procedures, Functions, Packages, Triggers. Programmatic SQL – Embedded SQL

Unit-4:

Concurrency Control: Transaction processing, locking techniques, database recovery, security and authorization. Overview of recovery techniques and Database Security. Overview of Object-Oriented concepts & characteristics ,Objects, OIDs and reference types, Database design for ORDBMS, Comparing RDBMS, OODBMS & ORDBMS

Unit-5:

Spatial data management ,Web based systems Overview of client server architecture, Databases and web architecture, N-tier Architecture, Business logic – SOAP , Multimedia databases , Mobile database Distributed Databases Introduction, DDBMS architectures ,Homogeneous and Heterogeneous Databases , Distributed data storage, Distributed transactions , Commit protocols , Concurrency control & recovery in distributed databases , Directory systems

Suggested Books and References

Text Books :

1. Silberschatz Abraham, Korth Henry & Sudarshan S., Database Systems Concepts, McGraw Hill, 1997.
2. Date C.J., an Introduction to Database Systems, Addison Wiley.

Reference Books

1. Bipin Desai, An Introduction to Database Systems, Galgotia Publications, 1991.

Suggested E-resources

1. www.javatpoint.com
2. www.tutorialspoint.com

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CAL815 Web Technology 4 Credits (3-0-2)

Syllabus

Unit-1:

Introduction to PHP, History of PHP, Installation and configuration of PHP, PHP Variables, Scope of variables, PHP String, string manipulation, PHP Operators, Precedence of Operators, Expressions. Basic HTML, PHP \$_GET, PHP \$_POST.

Unit-2:

Conditional Statements, if, PHP If...Else, nested if... else..., PHP Switch, PHP Looping, PHP Functions, Built in and user defined function, regular expression functions, Math, date and time functions, PHP Arrays.

Unit-3:

MySQL Connect, MySQL Create, MySQL Insert, MySQL Select, MySQL Where, MySQL Order By, MySQL Update, MySQL Delete, PHP ODBC.

Unit-4:

Input validation, PHP with client side scripting language PHP Error, PHP Exception Cookies and Session, Code reuse, require(), include(), and the include path, file system functions and file input and output, file uploads Use of CSS

Unit-5:

Introduction to Word Press and its usages ,Word Press Installation, A Quick Tour of Word Press Dashboard and its working interface, Building a Website using with Word Press Dashboard and Theme, Installation of Plugin and Themes with Word Press and Making Navigation and Page, Using Word Press Plugin SEO,Contact Form, Social Plugins, Post & Pages,Uploading Web site on Web Server, Introduction to Woocommerce, Woocommerce Installation, Working on Woocommerce using with tools.

Text Books:

1. Atkinson, Leon. Core PHP Programming, New York: Prentice Hall.
2. David Sklar, Learning PHP 5, O'Reilly Media
3. Charles, Mastering PHP 4. 1 BPB Publications
4. Lisa Sims, Building Your Online Store with WordPress and Woocommerce: Learn to Leverage the Critical Role, Apress

References Books:

1. Expert PHP and MySQL, Wrox Programmer to Programmer, Wrox Press, 2010
2. PHP for Absolute Beginners, 9781430224730 (1430224738), Apress, 2009
3. Adrian W. West & Steve Prettyman, Practical PHP 7, MySQL 8, and MariaDB Website Databases: A Simplified Approach to Developing, Apress,2018
4. Dwight Robert, PHP: Learn PHP in 24 Hours or Less - A Beginner's Guide To Learning PHP Programming Now, CreateSpace Independent Publishing Platform,2016
5. Robbert Ravensbergen, Building E-Commerce Solutions with WooCommerce, 2nd Edition: Transform your WordPress website, Packt Publishing Limited,2015

Suggested e resources

1. <https://www.tutorialspoint.com/php/>
2. <https://www.tutorialspoint.com/html/>
3. <https://www.tutorialspoint.com/wordpress/>
4. <https://kinsta.com/blog/woocommerce-tutorial/>
5. https://www.nptelvideos.com/php/php_video_tutorials.php

CAL818 Cloud Computing 3 Credits (3-0-0)

Syllabus

Unit-1:

Cloud Computing: Existing usage of cloud computing; New paradigm in the cloud; Applications. Cloud Computing Architectural Framework: Cloud: Benefits, Vocabulary, Business scenarios, Essential characteristics, Deployment models, Service models, Multi-tenancy, Approaches to create a barrier between the tenants.

Unit-2:

Vendor Lock-in and Efforts at Standardization: Need of migration; Preventing vendor locking, Comparison chart., Data Center Operations: The anatomy of cloud infrastructure, Data Center Operations, Security challenge, implements "Five

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Principal Characteristics of Cloud Computing, Data center Security Recommendations. Introducing Identity Services, Enterprise Architecture with IDaaS , IDaaS Security Recommendations, data Security in Cloud, technologies for data security.

Unit-3:

Governance and Enterprise Risk Management: Information security governance processes, Governance and enterprise risk management in Cloud Computing, Governance Recommendations, Enterprise Risk Management Recommendations, Information Risk Management Recommendations and Third Party Management Recommendations

Unit-4:

Cloud Reliability, Fault Tolerance and Response Time: Business continuity management: System reliability, Case studies on designing for reliability; Concept of fault tolerance; Response time. Internet Cloud Security: Introduction; Potential threats; Security as a service by cloud providers; Fraud theory and Intellectual property.

Information Lifecycle Management: Key challenges regarding data lifecycle security, Data Security Recommendations by Cloud Computing

Unit-5:

Traditional Security, Business Continuity, and Disaster Recovery: Risk of insider abuse, Security baseline, Customers actions, Contract, Documentation, Recovery Time Objectives (RTOs), Customers responsibility Case Studies : Amazon's cloud services (AWS)

List of Experiments : N/A

Suggested Books and References

Text Books :

1. David, E.Y. Sarna, *Implementing and Developing Cloud Computing Applications*, CRC Press.
2. Dimitris, N. Chorafas, *Cloud Computing Strategies*, CRC Press.
3. Rajkumar Buyya, James Broberg, Andrzej M. Goscinsk, *Cloud Computing: Principles and Paradigms*, Wiley Publications

Reference Books

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1. Mather, T., *Cloud Security and Privacy: An Enterprise Perspective On Risks And Compliance*, O'Reilly

Suggested E-resources

1. [https://mrcet.com/downloads/digital_notes/IT/CLOUD%20COMPUTING%20DIGITAL%20NOTES%20\(R18A0523\).pdf](https://mrcet.com/downloads/digital_notes/IT/CLOUD%20COMPUTING%20DIGITAL%20NOTES%20(R18A0523).pdf)
2. https://www.lpude.in/SLMs/Master%20of%20Computer%20Applications/Sem_2/DECAP470_CLOUD_COMPUTING.pdf
3. https://www.tutorialspoint.com/cloud_computing/cloud_computing_tutorial.pdf

CAL818 Programming with Python 4 Credits (3-0-2)

Syllabus

Unit-I

Introduction History, Features, Setting up path, Working with Python, Basic Syntax ,Variable and Data Types , Operator Conditional Statements If ,If- else ,Nested if-else Looping For, While ,Nested loops Control Statements Break, Continue

Unit-II

String Manipulation Accessing Strings ,Basic Operations ,String slices ,Function and Methods Lists Introduction ,Accessing list ,Operations ,Working with lists ,Function and Methods Tuple Introduction ,Accessing tuples ,Operations ,Working ,Functions and Methods

Unit-III

Dictionaries Introduction, Accessing values in dictionaries ,Working with dictionaries ,Properties ,Functions Functions Defining a function , Calling a function, Types of functions ,Function Arguments ,Anonymous functions ,Global and local variables

Unit-IV

Modules Importing module ,Math module ,Random module ,Packages ,Composition InputOutput Printing on screen ,Reading data from keyboard ,Opening and closing file ,Reading and writing files ,Functions

Unit-V

Exception Handling ,Exception Handling
,Except clause ,Try ? finally clause ,User Defined
Exceptions OOPs concept Class and object , Attributes
,Inheritance ,Overloading ,Overriding ,Data hiding

Suggested Books and References

Text Books :

1. Learning Python by Mark Lutz, David Ascher
Shop O'Reilly - O'Reilly Media
2. Beginning Python Magnus Lie Hetland ,
Goodreads
3. Python Programming for the Absolute Beginner
third edition Ross Dawson Goodreads

Reference Books

1. Practical Programming: An introduction to
Computer Science Using Python, second edition,
Paul Gries, Jennifer Campbell, Jason Montojo,
The Pragmatic Bookshelf.
2. Python for Informatics: Exploring Information,
Charles Severance
3. Learning Python, Fourth Edition, Mark Lutz,
O'Reilly publication
4. Introduction to Python for Computational
Science and Engineering (A beginner's guide),
Hans Fangohr